



VMRS Chatbot Dec 2025

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The Problem

- Background
 - VMRS (Vehicle Maintenance Reporting Standards) is an industry standard to categorize parts, but individual vendors may categorize the same part differently.
 - 002-010-084 (system, assembly, component)
 - Reyes Holdings Fleet currently has 30,000 parts and utilized ~19,000 parts in 2025.
- Challenges
 - Without correct mapping of vendor part VMRS codes, we cannot effectively identify where optimizations can be made
 - Fleet management team must manually find VMRS codes and may not be 100% accurate.

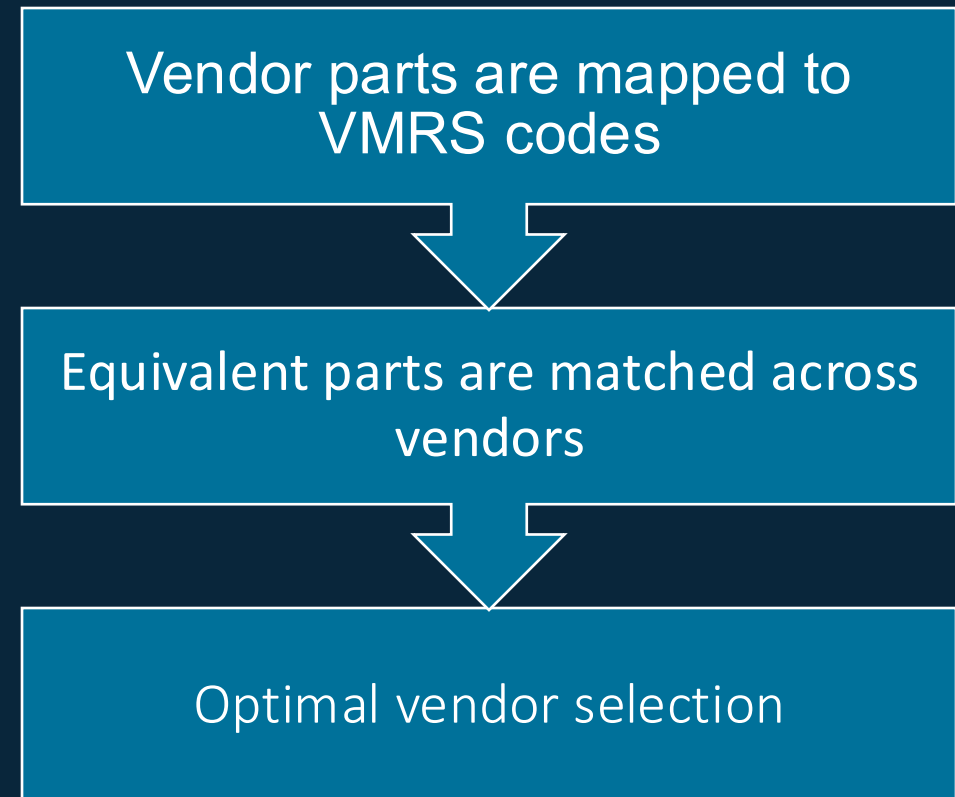
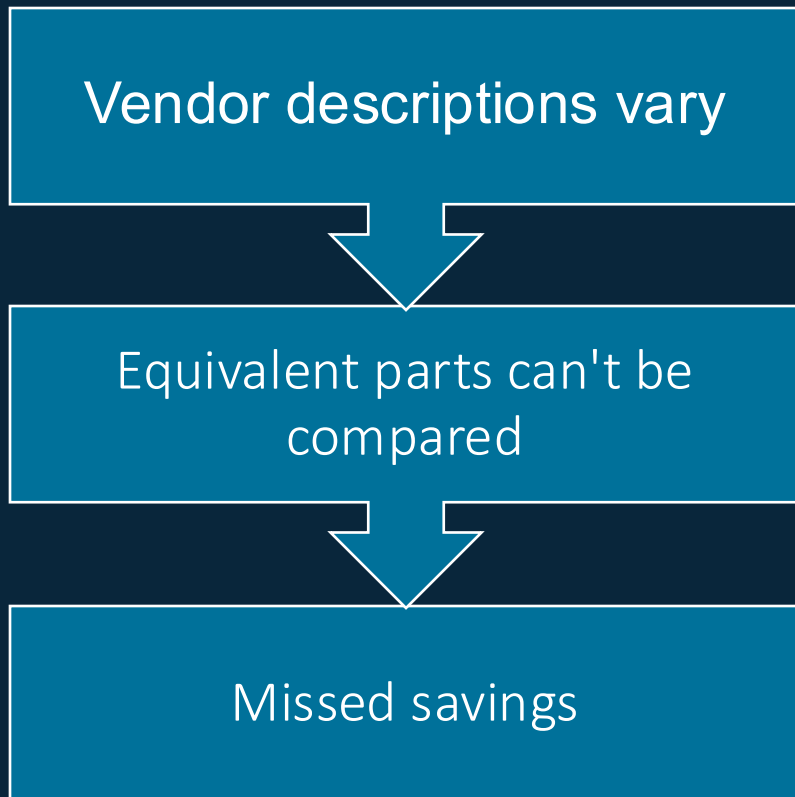
Demo

- What is the VMRS code for grille?
- What is the VMRS code for flux capacitor?

Savings and Cost

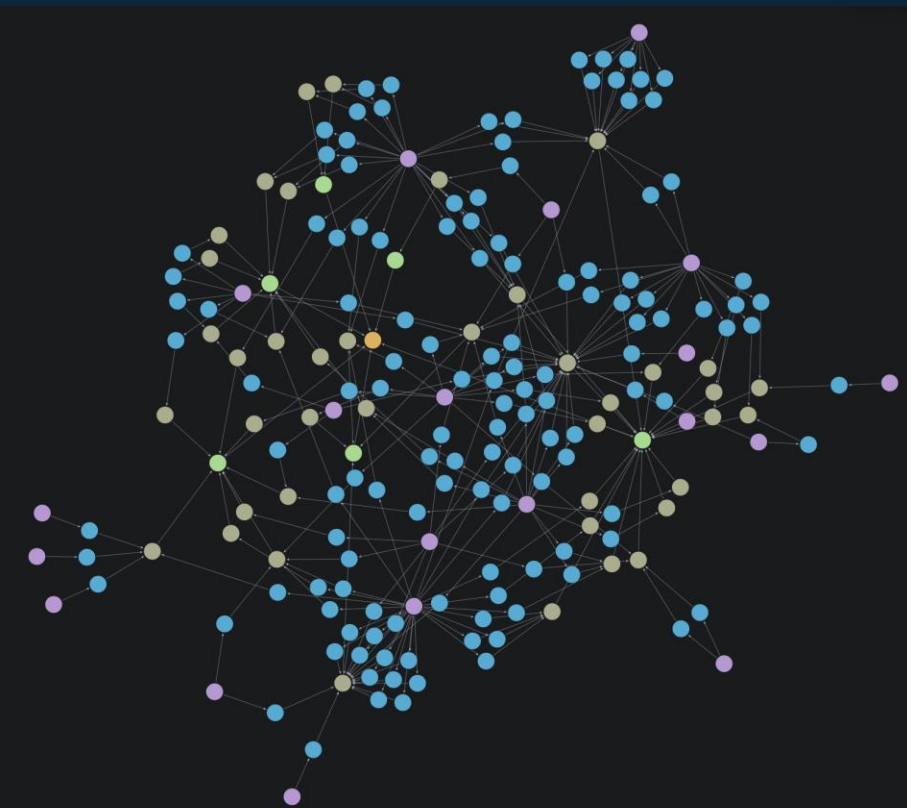
- Previous Initiative
 - Alternative Filter Vendors Annual Savings: ~\$160,000
- Current Labor Annual Cost: \$75,000
 - 5,000 parts * .5 hours per part * \$30 per hour
- Parallel Costs
 - Data Cleanup One Time Fee: ~\$20,000
 - Data Cleanup Annual Cost: \$5,000
 - \$1,250 quarterly
 - Expanded coverage: 240 vs 19,000 parts

Current Problem vs New Solution



The Approach

1. Acquire Source of Truth: Official VMRS handbook
2. Ingest handbook and vendor data into knowledge graph



Nodes (201)

* (201)

Assembly (6)

Component (41)

System (1)

Vendor (20)

VendorPart (133)

```
1 MATCH p=(n:System)-[]-()-[]-()-[]-()-[]-()
2 WHERE n.code = "022"
3 RETURN p
```

Results and Limitations

- Solution confidently predicts system and assembly codes.
 - Component requires user feedback and more data.
- Solution does not have access to pricing information.
 - Snowflake read access is required.

Tech Stack

- Knowledge Graph: Neo4j was selected for generous free tier
 - Any knowledge graph provider is interchangeable
- LLM: Claude was selected for llm and interface in one subscription
 - Any LLM is interchangeable (Copilot)



Next Steps

- Roll out pilot to Fleet Management team
- Learn from Fleet Management operators
- Get Snowflake read access

Appendix

Savings Details

- Purchase Optimization: One previous optimization over 240 parts allowed \$160k in annual savings.
- Data cleanup cost: 1 time fee of ~\$20,000 for outsourced VMRS code cleanup to 3rd party company and is not 100% accurate, with an additional \$500 to \$1,250 quarterly costs.
- Up to \$18.75k quarterly Labor Cost: Quarterly, 1,250 parts require 15-30 minutes each to research/validate, labor hrs. 312-625, labor cost and ~\$9,400 - ~\$18,750

Appendix 2: Development process including validation

1. Extract entities and relationships from official VMRS handbook.
2. Combine extracted data with corrected vendor information.
3. Upload combined data to knowledge graph

Appendix #3

- Link to handbook: [VMRS Industry knowledge](#)